

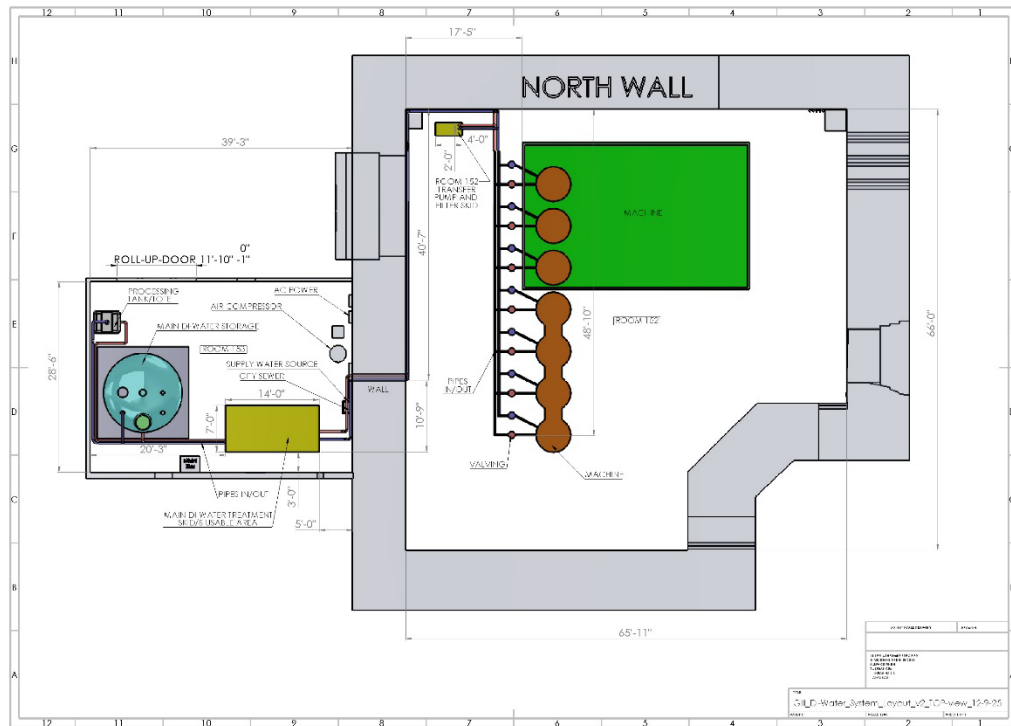
# **Gamble III (GIII) De-Ionized Water System Requirements**

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NRL Code 6770

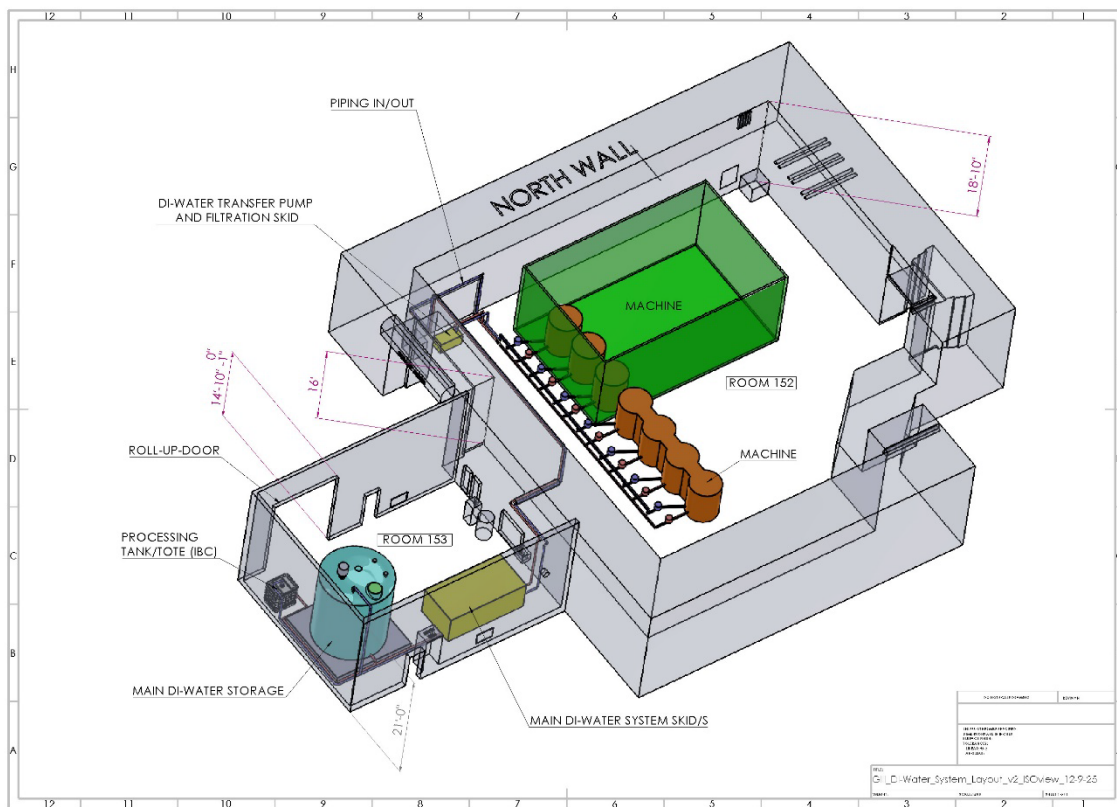
With assistance from:  
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**SCOPE OF WORK:**

1. Provide complete system costing with and without installation of piping, water storage, and a pure Deionized (DI) Water Generating System including the Pretreatment Water Filtration and Conditioning Equipment, Reverse Osmosis Membrane System, Deionization Equipment, Distribution Pumping Equipment and Advanced programmable Logic System Controls with programs that will provide completely automated system functionality, continuous dashboard style display of all performance values, reactive alarm handling for minimal Operator involvement, and mirrored Operator system control in multiple rooms. This includes system performance data logging for after-the-fact verification of continuous water quality, flow, pressure, etc. and be capable of communication with Ignition software by Inductive Automation, an Ethernet based Facility Automation System, to meet the requirements set forth in this proposed specification document.
2. The Gamble III machine requires a liquid electrical insulator to be easily loaded into the process equipment. Absolutely pure Deionized Water provides an ideal medium to be used as an insulator in this application. The system needs to reliably produce very high-quality water that is devoid of ionic, particulate, transformer oil, and organic contamination which is ideal for use as an insulator. The DI Water is easily filled into the machine and can be evacuated from the machine when required to either refresh the water quality or when servicing of the machine is required.
3. The system shall include Hydrocarbon Adsorption of the evacuated water to remove transformer oil that may emanate from the machine and become entrained in the evacuated DI Water. Source Water for the DI water Generating System shall be non-potable chlorinated municipal water. The system is designed to purify the source feed water to provide a highly resistive liquid medium in the Gamble III machine.
4. The Deionized Water Production equipment shall be completely factory assembled on a single skid or on multiple interlocking skids as convenient for assembly and shipping. The system requires reduced field preparation and installation using a forklift. The mixed bed resin bottles are to be located on the provided concrete slab floor of NRL Building 256 Room 153, easily removable from the system using CPVC unions, and easily accessible for bottle-cart change-out. The DI-Water filtration system skid/s will also be located on a provided monolithic concrete pad at ground level in Building 256 Room 153 approximately 300-linear-feet from the Gamble III machine which is in another room (NRL Building 256 Room 152) also at ground level on a monolithic concrete pad floor. The approximate 600-linear-feet of 3-inch CPVC in/out piping and air compressor piping will have to go through an 8-foot-thick concrete wall which has a pre-drilled 12-inch diameter hole, up to a height of 17-feet, back down to ground level, and around two corners to tie into the machine using 3-inch CPVC unions, 3-inch CPVC ANSI flanges, and 3-inch NPT CPVC fittings. The in/out plumbing and electrical conduit feeds for the main system skid/s shall be installed on the floor over to the East and South walls of Room 153 at a height of no more than 1-foot so the system is accessible on all sides. The system will require two 300gpm variable frequency drive transfer pumps, one in Room 152 and one in Room 153, to pump the DI-Water over a wall to/from the machine, see figures 1 and 2. The machine will require controlled valves for the in/outs per the schematic on the last page of this document.

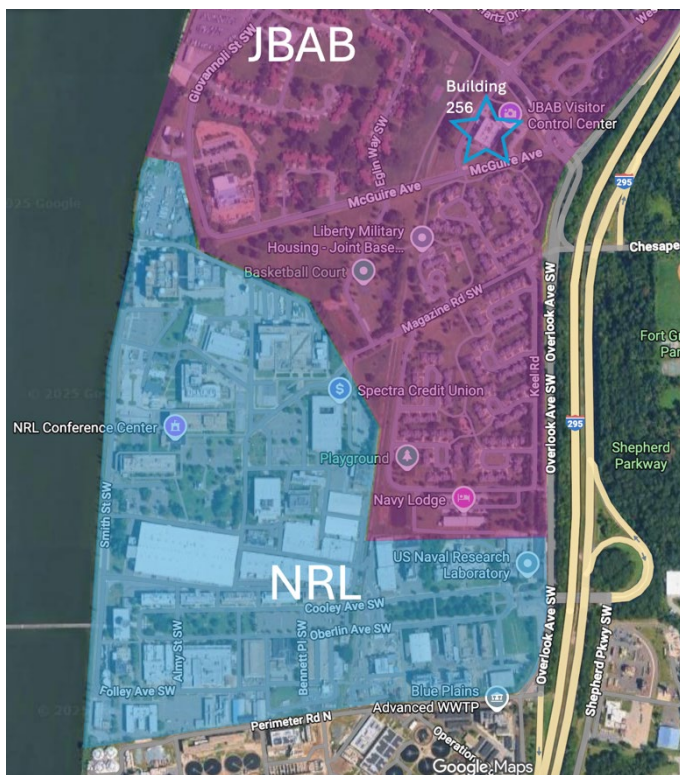


**Figure 1 – Gamble III Facility Top-Down View:** Showing the overall DI-Water system configuration, sizing and piping on JBAB in NRL Building 256 Room 152 and Room 153.



**Figure 2 – Gamble III Facility Isometric View:** Showing the overall DI-Water system configuration, heights, and piping on JBAB in NRL Building 256 Room 152 and Room 153.

5. The Deionized Water Production equipment will have on-site storage in Room 153 via a main 10,000-gallon storage tank for storing DI-Water and/or polishing the water via a kidney loop to the main DI-water system. The system will also have a second smaller 275-300-gallon buffer tank/tote for evaluating and processing DI-water that has been potentially contaminated with oil or particulate during a catastrophic event in the machine where large amounts of particulate and/or transformer oil are introduced into the DI-Water. This smaller processing tank provides a location to send the potentially very dirty water to for evaluation and inspection. This process will be done manually and may require the DI-Water from the machine to be inspected a few 100-gallons at a time. From the smaller tank/tote the DI-water can be either cleaned and/or put back into the main DI-Water storage tank, disposed of, or forklifted to an on-site oil purification system for transformer oil reclamation.
6. An onsite visit scheduled with the NRL team, see Figure-3, is mandatory for all potential manufacturer vendors, prime bidders, or general contractors prior to submission of bid to inspect the installation site requirements and to confirm acceptability of the provided space, power, supply water, and sewer utilities. Any vendors, bidders, contractors (including sub-contractors) installation personnel must be U.S. Citizens in order to access NRL and JBAB.



**Figure 3 – NRL Building 256 Location:** For our purposes, the address for NRL Building 256 is U.S. Naval Research Laboratory, 4555 Overlook Ave SW, Building 256, Washington, DC 20375. However, the physical address for building 256 is 256 McGuire Ave, Washington, DC 20032, as it is within the JBAB fence line and shared with the JBAB Visitor Center.

7. The facility may provide: rest rooms, a NRL forklift to unload and position hardware as needed, limited uncovered outdoor staging, limited covered staging in Room 153.

8. **Installation:** The main filtering skid and secondary transfer skid hardware for this system are to arrive at NRL **12-months to 18-months** after the contract award. The delivery of the storage tanks to the facility on JBAB at NRL Building 256 and the final facility installation and commissioning for testing and validation are to occur **12 to 36 months** after the contract award.
  - a. **Acceptance Criteria:** Proof of system operation with 7 gallon per minute DI-Water production at 18-MO $\Omega$ -cm Resistivity and complete control and automation of system hardware including, but not limited to: VFD transfer pumps, valving, filtration, diagnostics, historization of data, programming processes, and control with Ignition software to ensure system operation and installation in NRL Building 256 per the requirements stated in this document.
9. **Warranty:** 1-year on all parts and service starting after system acceptance criteria have been met. The manufacturer must provide a call-back response time by phone of less than **1-business day** and on-site service **with-in 2-business** days to facilitate and support operations.
10. **Progress payment options:** This information will have to come from an NRL purchasing agent.
11. **Inflation Approach:** Dependent on contractor

#### **SYSTEM DESIGN OPERATING PARAMETER REQUIREMENTS:**

1. Minimum Deionized Water Production Rate: 7 Gallons Per Minute (gpm) with 10 gpm feed city water
2. Design DI Water Transfer Flow: 300 gpm
3. Minimum Pure Water Storage Volume: 10,000 Gallons
4. Water Quality: Greater than 18-MO $\Omega$ -cm Resistivity on 8000 gallons of city water in less than 24 hours
5. Operation: Automatic, PLC communicates and controlled with facility software Ignition by Inductive Automation via Ethernet cable. Historization is required for total organic carbon, resistivity, particulate count, %-dissolved gas, pump speeds, differential pressures across filter vessels and pumps, and energy use
6. A single PLC with distributed I/O is acceptable and programmed to run procedures in Figure-4 system schematic and stated in this document.
7. Control cabinets must have manual emergency stop buttons and lock-out-tag-out disconnects within 8-feet of each pump.
8. Built on a skid/s-based system, installation in Indoor mechanical space on monolithic concrete slab (provided), must be assembled by manufacturer, positioned on-site, no crane provided in Room 153, use of a forklift approved if need be.
9. Provided concrete slab floor space for the DI-Water main purification system skid/s in Room 153 is 7-foot by 14-foot, maximum skid assembly height of 8-feet, the storage tanks in Room 153 will have their own space and will not take up any of the 7-foot by 14-foot floor space requirement of the skid/s. See Figures-1&2
10. Provided concrete slab floor space for the DI-Water secondary transfer/filter skid in Room 152 is 2-foot by 4-foot, maximum skid assembly height of 8-feet, See Figures-1&2

11. Available power in the main Room 153 for DI-Water polishing/filter system, controls, diagnostics, air compressor, and first transfer pump: 277/480volt-60amps-3Pole, 208/240volt-60amps-2Pole, 120/208volt-40amps-1Pole
12. Available power in machine room 152 for DI-Water controls, diagnostics, and second transfer pump: 208/240volt-40amps-2Pole, 120/208volt-20amps-1Pole
13. Power to system must be supplied along the floor via. conduit, maximum conduit height of 1-foot, from East wall of Room 153
14. Must operate on building/city supplied 1.5-inch diameter water supply line at 40psi-dynamic/50psi-static pressure or the system will require and need to include a booster pump to meet manufacturers specifications.
15. Building supplied sanitary sewer pipe is 3-inch diameter at atmospheric pressure
16. Maximum flow to city sewer for discharge is 100gpm
17. Two 300gpm variable frequency drive (VFD) transfer pumps, one in each of two rooms (Room 152 and Room 153) to pump DI-water up and over a 17-foot door/wall height
18. Valves to be pneumatically driven and manual control per requirements in Figure-4 schematic on last page of this document.
19. Provide a dedicated air compressor for compressed air with sufficient pressure and flow for all necessary DI-Water system's valves located in Rooms 152 and 153. This includes approximately 300-feet of installed pneumatic/system-rated pipe.
20. Reverse Osmosis membrane operating pressure of 150-225psi and an acceptable feed water pressure of 40psi dynamic.
21. Electronic Deionization with automatic switching DC Power Supply and on-screen voltage control. 90-92% recovery, 8000 gallons in 24 hrs.
22. Automatic Water Softener – Duplex: Less than 1 grain per gallon (gpg) Total Hardness
23. De-bubbler, sized for less than 70% dissolved gas (measured with an O2 sensor)
24. Ultraviolet-C (UVC) Sterilization: Single Unit Dual Wavelengths, Primary wavelengths of 185nm and 254nm, wattage and number of bulbs based on 50gpm flow and less than 200ppb of organic carbon, 316 or 304 stainless steel outer reaction chamber with pure quartz tube isolation between UVC lamp/s and process water. Elastomers in contact with pure water shall be of High Temperature Silicone, EPDM or VITON. Operator Interface shall display UVC lamp status, a cumulative Run Time hour meter and shall output a service alert for lamp replacement whether due to a lamp failure or elapsed run time.
25. Delivery pressure to machine and storage tanks must be bellow 150psi
26. Total Organic Carbon: Less than 200 parts per billion (ppb)
27. Chloride: Less than 50ppb
28. Silica: Less than 500ppb
29. Free Chlorine: Less than 0.15 parts per million (ppm)
30. Particulate filtration: 1 Micron with Beta 1000 efficiency at 300gpm
31. Oil filtration: activated carbon for up to 300gpm and at least 90% efficiency on a single pass. OAM2 in Figure-4 is up to 50gpm.
32. Digital diagnostics for state, status, historization, and error monitoring include a particulate counter, oxygen sensor, total organic carbon sensor, resistivity, pressure, flow, level sensors on storage tank and processing equipment
33. Analog diagnostics on equipment include pressure and flow
34. One-piece vertical 10,000-gallon high-density cross-linked Polyethylene storage tank cylindrical in cross section, virgin material construction rotationally molded, Must use Specific-Gravity greater than or equal to 1.8 for storage tank design and provide wall thickness



- mapping, integrally molded flanges, fully drainable via gravity, positioned safely on a provided flat 10-inch tall concrete slab, integrally molded flanges/ports, and include a level sensor and flow shut-off, installation sizing for rigging and placement need to be determined by bidder.
35. Storage tank ports must include: one ANSI Flange for 3-inch CPVC pipe on bottom ported out the side for full draining capability, one ANSI Flange on top for 3-inch CPVC pipe that extends down into the tank 4-feet for recirculate. One ANSI Flange for 3-inch CPVC pipe on top for 300gpm 5micron filtered venting to atmosphere (include tank vent), One ANSI Flange or equivalent for level sensor, One access hatch/port for personal to do pressure wash cleaning.
  36. Provide three 275-300-gallon Intermediate Bulk Containers (IBC) forklift ready totes, one tote piped into the system using 3-inch CPVC couplers located at the top and bottom, for easy removal from the system.
  37. Approximately 600-linear-feet of 3-inch CPVC: Pipe, NPT, and ANSI Flange fittings.
  38. Mixed/Resin Beds/Bottles to be accessible via bottle cart on North Side of main filtration skid to be brought in and out of Room 153 through the roll-up door.
  39. All piping to be sufficiently supported and/or secured on floor and/or walls in Room 152 and Room 153
  40. Piping or conduit to/from main purification system in Room 153 from East and/or South wall to be along floor, maximum height of 1-foot.
  41. Decontaminated water to go to city sewer will have a balanced pH between 6 and 9 with no free-floating transformer oil on/in it, dissolved transformer oil in the wastewater is acceptable.
  42. Provide literature, notes, and schematics of all hardware and final system design
  43. Supply three complete sets of spare consumables including: mixed/resin bed bottles, particulate filters, water softener media, sodium chloride solution, waste neutralizing media, and all carbon filters.

#### **DEIONIZED WATER PRODUCTION AND DELIVERY PROCESS DESCRIPTION:**

This system is designed to produce high purity water and is to be assembled from carefully selected water treatment technologies for the most efficient and cost effective operation available. This system is capable of operating on a wide variety of source water chemistries and will be capable of storing 10,000 gallons of purified water. Please refer to notes in this document regarding feed water and deionized water quality requirements. This system is composed of components for non-potable city water conditioning and filtration, reverse osmosis membrane purification, electronic deionization, water storage, water distribution and delivery pumping, ultraviolet sterilization and total organic carbon (TOC) reduction, ion-exchange deionization (DI) and particulate removal with micro-filtration, and de-gassing. The pretreatment equipment conditions, filters, and removes the dissolved salts from ordinary tap water. This pretreated water is then stored, further deionized, sterilized, filtered a final time, degassed, and continuously circulated from the storage tank. The finished water is available under pressure for supply to the machine and then returned to the polishing equipment and/or storage tank under pressure. Please refer to the following Process Sequence and schematic at the end of this document for the DI-water system design.

#### **DI WATER PRODUCTION AND DELIVERY PROCESS SEQUENCE:**

1. NOTE: The following sequence is the methodology and architecture of the design.
2. The System is supplied with chlorinated municipal non-potable tap water.

3. The raw feed water is first filtered by a permanent media, self-cleaning Media Filter. The Media Filter removes relatively large debris and finer particulate material from the source water to keep downstream equipment clean and to reduce any loading caused by source water particulate material on the subsequent filtration and purification stages of the DI Water generation process. The Media Filter automatically performs a backwash cycle to flush away any trapped particulate material. During Media Filter backwash, the Reverse Osmosis (RO) System is temporarily prevented from operating and then automatically enabled once the backwash cycle is completed.
4. The filtered water is then conditioned by the activated carbon filter. The carbon filter removes oxidizers, such as chlorine, and reduces organic molecules present in the source water. Chlorine, added to municipal water to help eliminate bacteria, is not tolerated by the reverse osmosis water purification membranes and cannot be allowed to pass into the reverse osmosis system. The activated carbon filter automatically performs a backwash cycle to reclassify the carbon filtration media and to flush away any trapped particulate material. During Carbon Filter backwash, the Reverse Osmosis System is temporarily prevented from operating and then automatically enabled once the backwash cycle is completed.
5. The carbon filtered water then flows through a Duplex automatic ion exchange water softener. The water softener continuously removes hardness ions (calcium, magnesium, iron) from the source water to protect the downstream reverse osmosis membranes from mineral deposit fouling. Hardness ions can reduce the life of the membranes and reduce the operational efficiency of the reverse osmosis system. The water softener media needs to be periodically flushed with a sodium chloride solution to remove the hardness ions it has collected during service. This process is called regeneration. Regeneration is performed automatically by the water softener according to the amount of raw feed water used by the system. The Reverse Osmosis System is temporarily prevented from operating and then automatically enabled once the backwash cycle is completed. All that is required for effective Water Softener operation is that an adequate level of water softener crystal salt is maintained in the water softener's brine production tank for use by the Water Softener during regeneration.
6. The conditioned water then flows through a 1 micron rated replaceable cartridge filter to remove fine particles that could clog the reverse osmosis membrane.
7. The conditioned and filtered water is then pumped to the operating pressure of the reverse osmosis membrane. At 150 – 225 psi, the membrane separates the dissolved ions (salts and minerals) and organic molecules from the pure water, flushing these contaminants to waste and sending the purified water to the water storage tank or the process equipment. The reverse osmosis membrane is typically capable of consistently removing at least 98% of the dissolved solids from the source water. The reverse osmosis system is activated and deactivated according to a set of level switch controls located in the pure water storage tank.
8. Trace ions remaining in the RO purified water are very efficiently and cost effectively removed at 98-99.6% contaminate rejection efficiency by in-line electronic deionization (EDI) then passed through a series and parallel arrangement of four deionization resin-beds to create polished deionized water flowing to the Pure Water Storage Tank. The system up to this point



is referred to as the pretreatment kidney loop that provides high purity water at the rate of the city water supply to save time and provide redundancy incase the second kidney/filtration-loop of the main system skid becomes contaminated.

9. When there is sufficient water volume in the pure water storage tank, the Deionized Water Delivery Pump can be activated, Purified water is then pumped from the pure water storage tank by the Delivery Pump and flows the water through the second kidney/filtration-loop purification polishing stages and is eventually delivered to the Gamble III machine or returned back to the DI Water Storage Tank via CPVC piping as a filtration/kidney loop.
10. One step in the purification polishing stage is that the water flows through a high intensity, low wavelength ultraviolet sterilizer which helps to control bacteria growth in the pure water and reduces organic content of the purified water through photo-oxidation.
11. Another step in the purification polishing stage is that the deionized water flows through Mixed Bed Ion Exchange Deionization Modules (DI Modules or resin beds) to ensure that the deionized water is free of ions and that carbon dioxide is continuously extracted from the purified water before being delivered to the Gamble III machine. DI Modules are essential in ensuring that the delivered water is at the highest resistivity to create an ideal liquid insulator within the machine.
12. The deionized water then flows through a Micro-Filter unit, equipped with 1-micron, absolute rated filter cartridges to prevent excursion of any very small particles and prevent any solids and ion exchange resin from traveling through the purified water delivery system.
13. The filtered deionized water then is degassed by a hydrophobic membrane that separates gas bubbles and dissolved gasses using a vacuum source as the driver for gas removal.
14. The purified water then flows out the supply port of the purification system and into the Facility deionized water delivery piping where the water can be injected into the Gamble III machine. The water then returns to the pure water storage tank from which it is continuously pumped through the polishing equipment and back out through the purified water piping network. Constant DI Water delivery pressure is maintained by an automatic back-pressure valve at the return point to the DI Water storage tank. This continuous circulation system configuration is designed to ensure that the deionized water continuously maintains its high quality.
15. When it is desired to flow water through the Gamble III machine, a 3-way flow selection valve is activated and DI Water flows from the DI Water system purification outlet to the machine.
16. When the machine purges used water, a variable frequency drive (VFD) evacuation/transfer pump, connected to a gravity pressurized drain port on the machine then pumps the used water from the machine either delivers the water to the large storage tank or removable collection tank/tote for inspection prior to discharging. The used water could then be processed by oil absorption and pH neutralization to be sent to the sewer for disposal.
17. The system's smaller processing tank/tote is for the inspection and processing of a 275-300-gallon volume that is either discharged directly from the machine using a branch installed tee

fitting with valve or from the bottom of the 10,000-gallon storage tank via an automatic/manual valve. The smaller storage tank/tote will be removable and will need to be outfitted with 3-inch quick disconnect CPVC unions. This tank/tote will be necessary for inspection and act as a buffer tank for evacuation of water in the event of a catastrophic failure. If too much oil gets into the DI-Water system processing equipment there will be a catastrophic consequence. The oil filter media is designed to claim at least 90% or more of the dissolved oil created during routine operation. In a catastrophic event where the transformer oil is no longer dissolved or entrained in the water it will be free to float to the top of the large storage tank, machine, or tote. If there is concern of a catastrophic event the machine would be required to sit idle to allow the oil to float to the top of the machine or storage tank where the 300gpm VFD pump would be manually turned on and valves set up to transfer the oily water and discharge it into the small tank/tote for inspection. If there is more oil than water the tote would be forklifted to the onsite oil purification system for reclamation of the oil. Once the free oil is removed from the tote the remaining water will be evaluated for reprocessing or to be sent to the sewer via the neutralization and waste management system described in this document.

18. The system shall be fully automated for unattended routine operation. The manufacturer of the system will provide advanced programmable logic controls (PLC), local touchscreen Operator Interface (OI) panels, and be controllable using Ignition software by Inductive Automation as the main facility control system from the operator control room. Since the DI-Water System will be isolated from the machine, a mirrored Operator Interface shall be provided for mounting within the machine operation Room 152 on the secondary transfer skid to enable system monitoring and control during DI-water equipment operation. System alarms, warnings, and maintenance alerts are all visually and audibly annunciated locally and at all control panels. The Advanced Controls are fully capable of being integrated into the local network for control, performance viewing, and alarm annunciation. The system will require programming of operations outlined in this document as well, but not limited to, program operations listed in Figure-3.
19. The OI panels shall display flow rates, pressures, storage tank and machine DI-water levels, resistivity, particulate count, % dissolved gas, and filter vessels differential pressures.
20. The DI-water system also needs manual emergency stop buttons and lock-out-tag-out power disconnects located on the main control panel of the main DI-water system skid in room 152 and within 6 feet of the second pump which is in Room 152 near the machine.

**The system configuration to meet the discussed system design, functionality, serviceability, control, and communication requirements to provide pure water service for the Gamble III machine will require:**

#### **1. Municipal Tap Water Treatment and Fresh Fill System**

- a. Automatic Multi-Media Filter
- b. Automatic Carbon Filter, Total Organic Carbon: Less than 200 parts per billion (ppb)
- c. Free Chlorine: Less than 0.15 parts per million (ppm)
- d. Silica: Less than 500ppb
- e. Automatic Water Softener – Duplex: Less than 1 grain per gallon (gpg) Total Hardness

- f. Reverse Osmosis Purification Unit - 8000 gallons, 98-99.6% contaminate rejection. in 24 hrs.
- g. Electronic Deionization – 90-92% recovery, 8000 gallons in 24 hrs.
- h. Chloride: Less than 50ppb
- i. Automatic Control Valves
- j. Water Quality Instrumentation for Pressure, Flow, and Total Organic Carbon.

## **2. Deionized Water Storage Tanks**

- a. Material: Molded Polyethylene, must use Specific-Gravity greater than or equal to 1.8 for 10,000 gallon storage tank design and provide wall thickness mapping.
- b. One Large Capacity: 10,000 Gallons completely drainable via gravity, must fit through a 11-foot\_10-inch tall by 14-foot\_10-inch-wide roll-up-door opening, the diameter of the tank must be less than 11-feet\_10-inches and the overall tank height including the base/stand and flanges must be less than 19-feet, it will have to be positioned/placed on a 10-inch tall concrete slab in Room 153. The room has a maximum usable height of 19-feet.
- c. Three Small Capacity: 275 - 300 Gallon forklift ready Intermediate Bulk Container (IBC) totes, one tied into system using unions, supply a quantity of three small capacity totes
- d. Continuous Water Level Instrumentation on large storage tank
- e. Large tank must have re-circulate capability from top of tank with minimal bubble generation (no cascading fill or return water streams) and fill/drain from bottom.
- f. Water Quality Instrumentation for resistivity, pressure, flow, and level
- g. Large tank access hatch for cleaning and inspection

## **3. Recirculating DI-Water Purification System**

- a. Organic Adsorption Module. Total Organic Carbon: Less than 200 parts per billion (ppb)
- b. Circulation Pump (45-50 gpm)
- c. Portable Exchange Type DI Resin/mixed bed Modules
- d. Ultraviolet Sterilizer Unit, total organic carbon less than 200ppb
- e. Micron particulate filter (1 micron Beta 1000) with monitored differential pressure and analog gauges
- f. Oil filtration: activated carbon up to 300-gpm and minimum of 90% oil removal on a single pass with new carbon, OAM2 in Figure-4 is up to 50gpm
- g. Flow Direction Control Valves
- h. Water Quality Instrumentation for Resistivity, Pump Pressure, particulate count, and resistivity

## **4. Mahine Fill and Degassing System**

- a. VFD Circulation Pump (300 gpm)
- b. Particulate Strainer (100 microns)
- c. Organic Adsorption Module, Total Organic Carbon: Less than 200 parts per billion (ppb)
- d. Micron Filter Unit (1 micron Beta 1000) with monitored differential pressure and analog gauges
- e. Degassing Module, less than 70% dissolved gas
- f. Vacuum Pump System for Degassing Module
- g. Flow Direction Control Valves
- h. Water Quality Instrumentation for dissolved gas, pressure, flow, continuous level, and resistivity

## **5. Water Return from Machine Purification System**

- a. VFD Transfer Pump (300 gpm)

- b. Particulate Strainer (100 micron)
- c. Organic Adsorption Module, Total Organic Carbon: Less than 200 parts per billion (ppb)
- d. Micron Filter (1 micron Beta 1000) with monitored differential pressure and analog gauges
- e. Flow Direction Control Valves
- f. Water Quality Instrumentation for pressure, flow, and total organic carbon.

## 6. Waste Treatment System

- a. Di-Water not to be aggressive to metallics using Solid Neutralizing Media and capable of adjusting pH to between 6 and 9 for up to 10,000 gallons
- b. pH monitor
- c. Rated for 100gpm

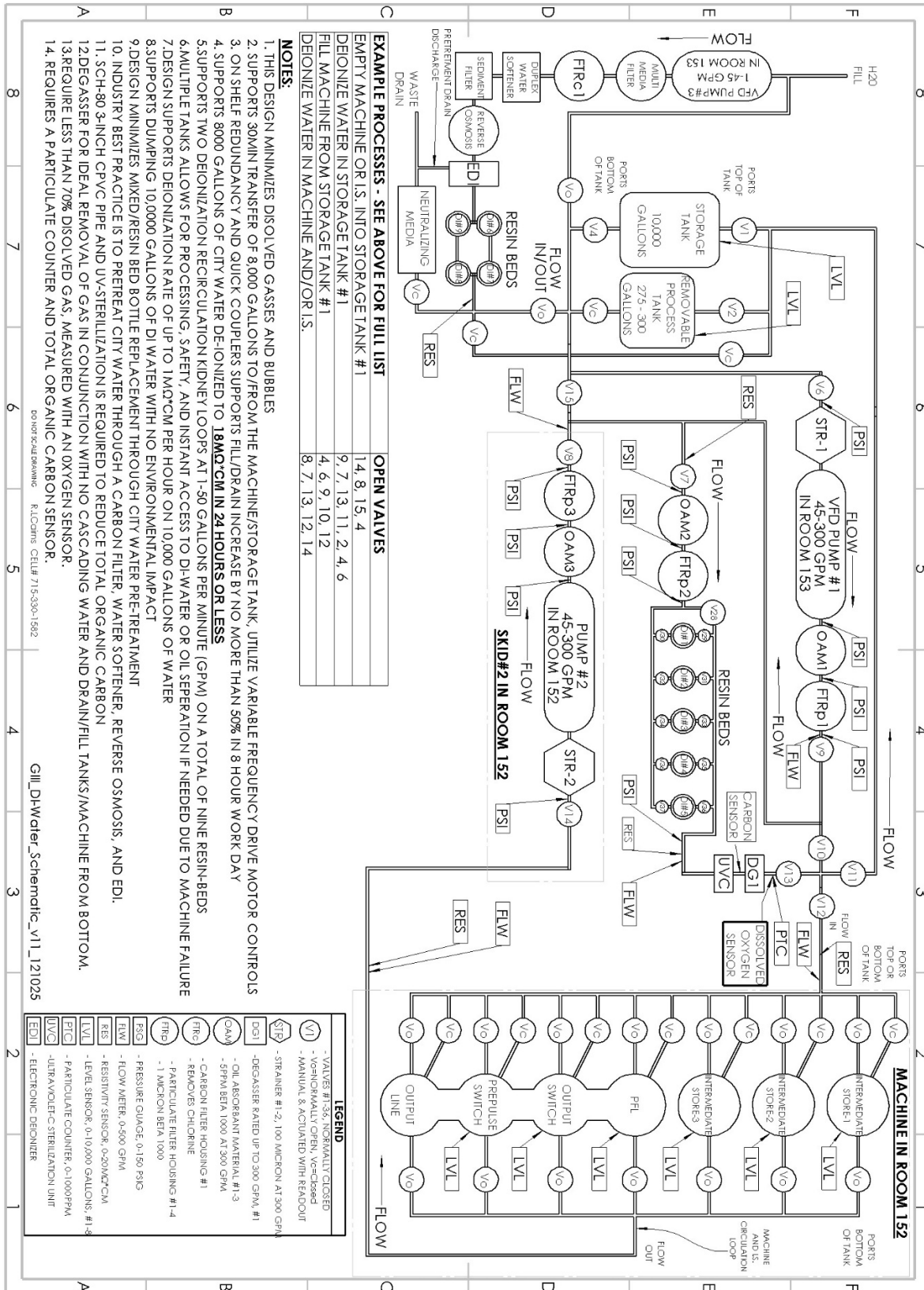
## 7. System Machine Controls

- a. Programmable Logic Controller controllable by Ignition software from Inductive Automation, used for facility controls.
- b. System Local Operator Interface and Mirrored Operator Interface in Room 153 and Room 152
- c. Automated Valving to be pneumatic with position sensing and local non-powered manual lever operation.
- d. For the valves on the machine that will experience significant EMP, use a 3-position switch (open, close, auto) located at the valve to control the air going to the actuator. Use a pressure switch on the airline going to the valve to determine the status of the valve. This eliminates the need to have any electrical sensor near the valve and allow the system to return to the failsafe position if needed.
- e. Provide Complete Programming of Operations without Cascading water to:
  - 1) Fill 10k-gallon-storage-tank using pre-treatment city supply line from bottom of tank
  - 2) Fill Machine from 10k-gallon-storage-tank, Fill/Drain using bottom ports
  - 3) Fill 10k-gallon-storage-tank from Machine, Fill/Drain using bottom ports
  - 4) Fill IBC-tote from Machine, Fill/Drain using bottom ports
  - 5) Fill Machine from IBC-tote, Fill/Drain using bottom ports
  - 6) Fill 10k-gallon-storage-tank from IBC-tote, Fill/Drain using bottom ports
  - 7) Drain 10k-gallon-storage-tank through wastewater treatment to city sewer, using bottom port
  - 8) Drain Machine through wastewater treatment to city sewer, using bottom port
  - 9) Drain IBC-tote through wastewater treatment to city sewer, using bottom port
  - 10) Circulate filter/polish 10k-gallon-storage-tank, using ports top to bottom
  - 11) Circulate filter/polish IBC-tote, using ports top to bottom
  - 12) Circulate filter/polish Machine, using ports top to bottom
- f. Provide Complete Programming of Machine Sub-Routine Operations: The Machine has a series of 7-parallel ports located on the top and bottom of each section of the Machine for cross flow, see Figure-4, and will require seven Machine circulation subroutines to circulate filter/polish these parallel ports individually top to bottom or in a combination of:
  - 1) Intermediate Stores#1, 2, & 3, top to bottom
  - 2) PFL. Output Switch, PrePulse Switch, & Output Line, top to bottom
  - 3) PFL top to Output Line bottom
  - 4) Output Line top to PFL bottom
- g. Independent Communication for Alarm Annunciation and Service Support

- h. Manual emergency shut off and lock-out-tag-out equipment within 8-feet of each pump

**8. Maintenance (Routine and Situationally Dependent)**

- a. Supply three complete sets of all major consumable products for the system including: mixed/resin beds, particulate filters, carbon filters, sodium chloride solution, neutralizing media, and water softener media.



**Figure 4 – Gamble III DI-Water System Schematic:** Overall layout of the DI-Water system including, but not limited to, all major water purification components, transfer hardware, diagnostics, piping, valves, programing information, requirements, and storage.